
Design Project 1 – End-Effector Design

ENGINEER IP13 – Integrated Cornerstone Design Projects in Engineering

Tutorial 07

Team THURS-12

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Submitted:

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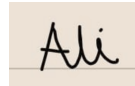
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Ali Ashmal Molwani

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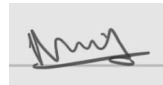


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Yaseen Nabag

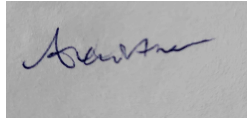
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(Student Signature) *

Executive Summary

Global logistics and warehousing operations face increasing pressure to process thousands of packages daily to sustain consumer demand. A critical bottleneck in this process is the manual handling of diverse, irregularly shaped, or fragile items, which contributes to slower operations and a high rate of workplace injuries, specifically manufacturing related serious injuries are about 41% above the average in Canada [1].

To address these challenges, we designed and prototyped a Robotic End Effector System specifically with the task of handling irregularly shaped objects in mind. The final solution features a double-jointed multi-claw mechanism mounted on a Q-Arm servo motor. The system utilizes a string-actuation method for uniform claw contraction, allowing it to adapt to various object while remaining under the strict 100g weight limit.

Main Body

Design Strategy

The design strategy prioritized adaptability, innovation and functionality.

Concept Selection: Our team had multiple iterations of different designs, including some “pincher” designs that made use of gears to pinch the objects to pick them up. Ultimately however, our team went with a multi-claw design because it was innovative, and we wanted to stand out from our peers by doing something more challenging and unique.

Actuation Method: A string-pull mechanism was chosen over a gear system. This choice significantly reduced the weight of the end effector, ensuring compliance with the 100g operational limit of the Q-Arm servo, and minimized friction points that could lead to mechanical failure.

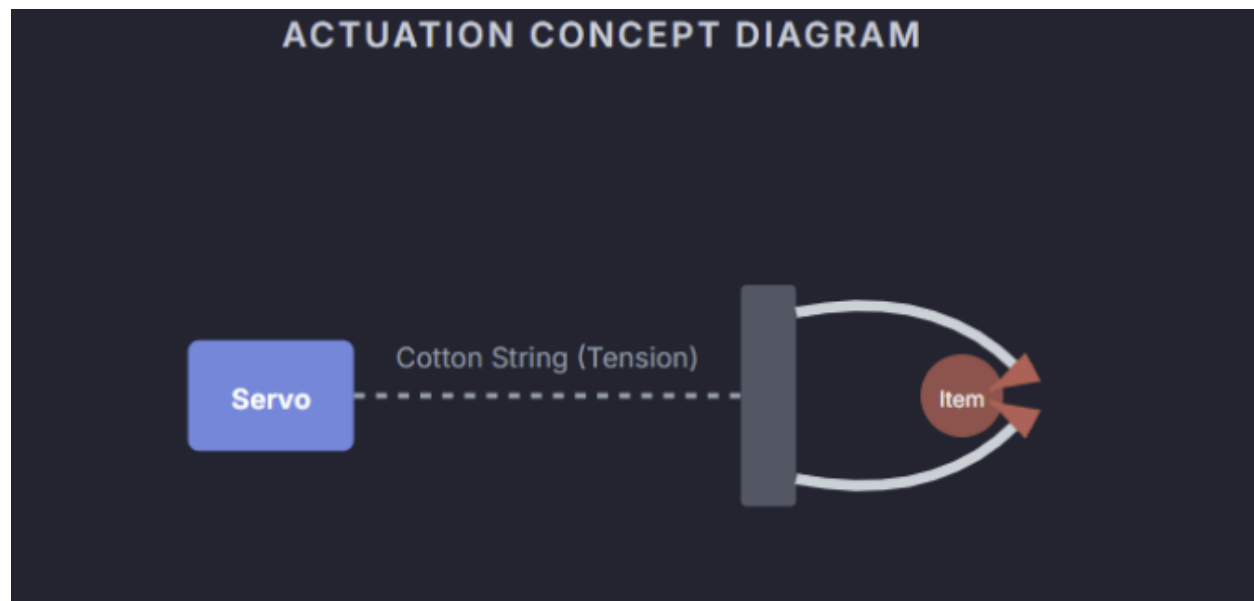


Figure 1: Visual descriptor of Actuation Concept

Material Optimization: To ensure sustainability and cost-effectiveness, the design utilized 3D printing for complex joint geometries while using standard fasteners for assembly (stove bolts, 3/16 X 1 1/2). The support structures were shortened to decrease the moment of inertia, thereby reducing the torque required by the motor to close the claws.

Software Implementation

The software component manages the entire warehouse workflow, from user authentication to physical sorting. The system was developed in Python, adhering to modular programming practices.

User security was a primary focus. Instead of storing plain-text passwords, the system utilizes the bcrypt library to salt and hash credentials [2]. This ensures that even if the users.csv database is compromised, user passwords remain protected.

Implementation of hardware and software was ensured using functions such as lookup products and pack products. The look up products function ensured that the scanned items exist in the inventory and return a price value. Then the pack products function uses a pre-determined coordinate system for each existing product and moves the Q-Arm to that location.

Iteration and Testing

For the software implementation, we tested our program for bugs and potential user errors. We scanned the barcode and did a test run to ensure that every function works as intended. Then, we did trial and error to move the Q-Arm to multiple locations until our arm arrived at every object. Then we used the arm's get coordinates function and implemented the list of collected coordinates into pack products, using the arm's set position function.

For the hardware implementation, we first made a prototype and realised that our secondary plate didn't have large enough holes to have it assembled properly. After fixing this issue we realised that our joints weren't moving correctly, hence we used bolts to add spacing to the joints, which allowed it to move in a predictable trajectory and reduce wear on the 3D-printed joints. Inserting metal bushings at the pivot points would reduce friction and increase the longevity of the mechanism.

We tested our mechanism extensively. It worked predictably when operated by hand; however, due to time constraints, we were unable to fully validate it using the Q-Arm during the final session (we were only able to get all objects till level 3 using Q-Arm). When operated by hand, the gripper was able to pick up objects from Level 1 to 4, except for the witch-hat. We believe that with additional time for calibration, the end-effector could successfully autonomously retrieve all 5 objects.

Reflection & Project Management

Comparing the actual project execution to the initial Gant chart shows that we have had to dedicate more time than initially expected in the execution of the project.

While the problem definition and software coding phases adhered closely to the schedule, the Fabrication phase required significantly more time than anticipated. This was primarily due to 3D printing times and the need to reprint components that required tolerance adjustments. Specifically, several initial prints failed due to a lack of implementation of supports in our GCode file. The tolerance adjustments for certain holes on our parts consumed extra time and materials, creating a bottleneck that reduced the time available for final testing.

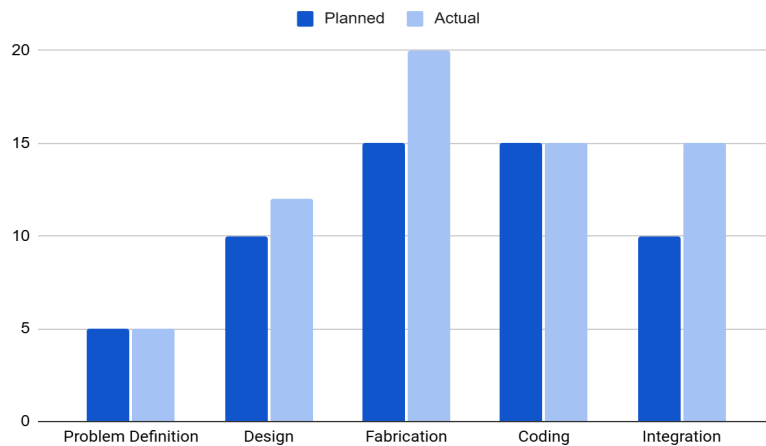


Figure 2: Comparison of Actual vs. Planned Number of Hours Spent on Phases of the Project

Refinements

If the project were to continue, the following refinements would be prioritized:

1. **Mechanical Torque Optimization:** The current string-pulley system relies heavily on the servo's direct torque. A gear reduction system could be implemented between the servo and the spool to increase gripping force, allowing the lifting of heavier objects.
2. **Dynamic Computer Vision:** Currently, the `pack_products` function relies on pre-determined coordinates for specific items. A future iteration would integrate a camera system to dynamically detect object coordinates, allowing the arm to pick up items regardless of their placement on the tray.

Reference List

- [1] "Manufacturing Planned Inspectional Initiative - WorkSafeBC," *Worksafebc.com*, 2022. <https://www.worksafebc.com/en/about-us/what-we-do/worksafebc-planned-inspectional-initiatives/manufacturing>
- [2] T. P. C. A. developers, "bcrypt: Modern password hashing for your software and your servers," *PyPI*, Feb. 28, 2025. <https://pypi.org/project/bcrypt/>
- [3] Ultimaker, "Ultimaker PLA Technical data sheet 1," Apr. 2022. Available: <https://um-support-files.ultimaker.com/materials/2.85mm/tds/PLA/Ultimaker-PLA-TDS-v5.00.pdf>

Appendices

Appendix A: Project Schedule

Preliminary Grant Chart:

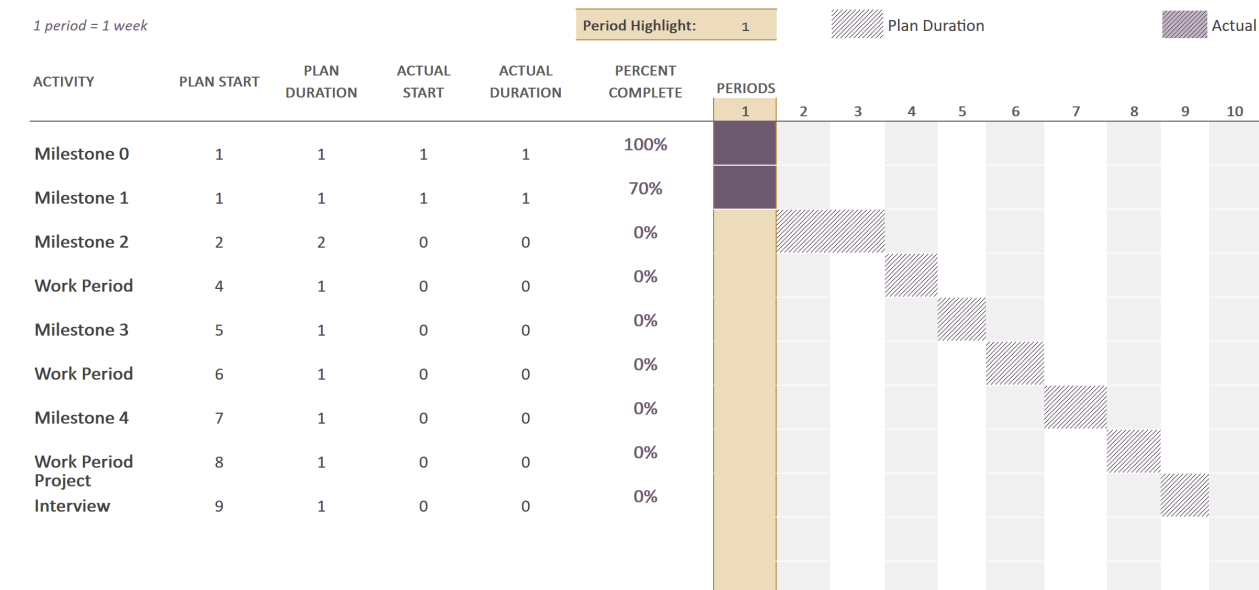


Fig 3: Preliminary Gantt Chart

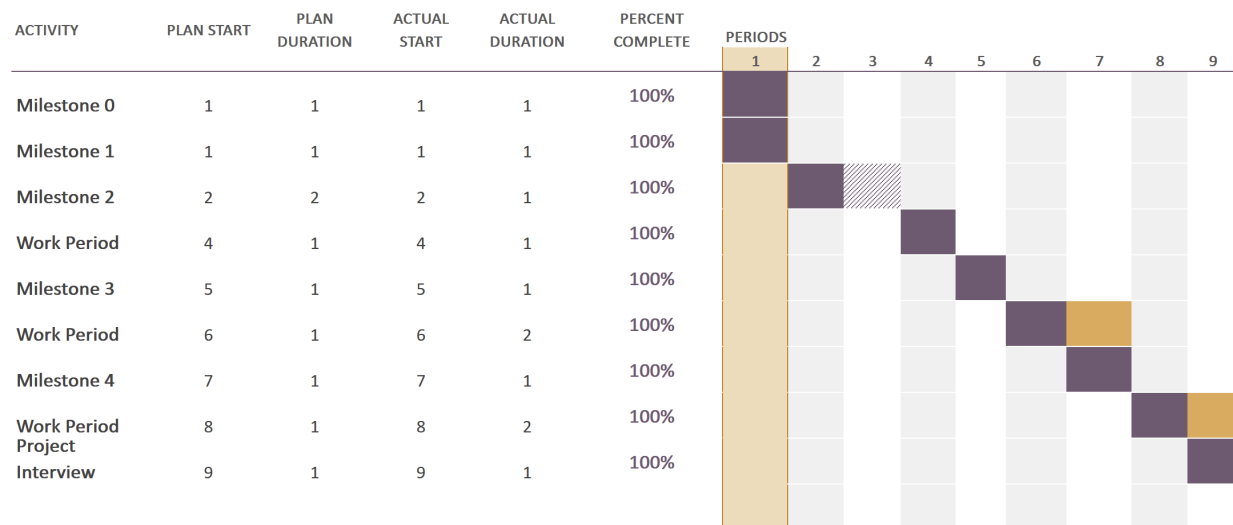


Fig 4: Final Gantt Chart

Date	Location	Topic & discussion	Duration
14th Nov 2025	Thode Makerspace	Coding and compiling the main program - The main task of this meeting was to mainly look over the team's individual codes and garner feedback from other team members. Then we compiled all our code into the team functions.	1hr 30 min
21st Nov 2025	Thode Makerspace	3D Printing the claws - After finishing our initial design, we got to prototyping. We did not get enough time in our design studio class, so we printed the remainder of our first prototype's part in Thode Makerspace.	1hr 30 min
26th Nov 2025	Thode Makerspace	3D printing the base - After our initial prototype, we determined a number of flaws with our design. First of the claw needed a support to connect to the motor, so we added that part. Additionally, the base required more weight, so we printed the base again with a heavier design.	1hr 30 min
28th Nov 2025	Thode Makerspace	3D printing longer joints - After initial testing, we found the torque provided by the motor was not enough to grip some of the objects consistently. To tackle this issue, we decided to print longer joints, thinking it would increase the torque, thereby increasing the grip strength.	1 hr 30 min

1st Dec 2025	PGCLL	Project report - We got together as a team to document all our progress and compile all our files.	1 hr
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Table 1: Additional Meeting Logbook

Appendix B: Scheduled Weekly Meetings

Date	Task
Week 5	- Complete milestones 0 and 1 - All team members start on their individual milestone 1. - Assign and review each team member a role. - Each team member starts an initial sketch for their design.
Week 6	- Complete milestone 3 - Finalise individual CAD designs and detailed sketches. - Choose a design to move forward with
Week 7	- Begin 3D printing and prototyping. - Continue making refinements to CAD design as necessary.
Week 8	- Continue 3D printing with refined designs - Review each function for the Q-arm and assign each team member a specific function - Work on flowcharts for each function
Week 9	- Work on each individual function - Begin thinking about the team functions and make a flowchart
Week 10	- Work on milestone 4 - Write the team functions for the Q-arm - Test the code and adjust as necessary

Week 11	- Assemble the final CAD design - Finalise the Q-arm code
Week 12	- Upload CAD design and project report - Interview - Final testing for all functionality

Table 2: Weekly Design Studio Agendas

Weekly Design Studio Meeting Minutes

Summary Week 5
Established team roles and analyzed the project scope. Completed the Team Charter and created preliminary Gantt Chart.
Date
Thursday October 7, 2025
Discussions
• Who is best suited for the Project Manager and Admin roles? • Analysis of Functions vs Objectives regarding worker safety and strain. • How to handle the constraint of lifting abnormally shaped packages.
Action items
• Submit Milestone 0 (Team Charter) and Preliminary Gantt Chart. • Refine Problem Statement to remove redundancies about package shapes. • Complete Milestone 1 (Objectives & Constraints). • Start working on Individual CAD models.

Summary Week 6
Brainstormed mechanism types. Debated between a gear-based pincher system and a string-actuated claw system.
Date
Thursday October 21, 2025
Discussions

• Whose Individual CAD model is the best • Can we somehow incorporate multiple CAD designs to create an even better design
• Pincher (gears) vs multi-claw (string) • How to ensure the claw closes uniformly on different sized objects.
Action items
• Begin CAD modeling of the chosen string-pull concept.

Summary Week 7
Finalized the 3D model for Milestone 2,
Date
Thursday October 28, 2025
Discussions
• How to fully constrain each part • How to incorporate the end effector platform into our CAD model • Timeline on when we should start printing a physical prototype of our design
Action items
• Submit Milestone 2 (Mechanism Decision & CAD). • Ensure all parts are fully constrained in the assembly file. • Export GCode for the initial test print.

Summary Week 8
Divided coding tasks among the team using a modular approach and started printing CAD model
Date
Thursday November 4, 2025
Discussions
• Assigning Python functions to members. • Division of work on the Team functions

• Which software to use for flowchart design • How to reduce print time • Why did our wheel print fail
Action items
• Individual members to start coding their assigned Python functions. • Create flowcharts for pack_products and main functions (Milestone 3). • Create Individual flowcharts • Break 3D model into more parts so that it takes less time to individually print each part • Add supports to the Print next time to prevent it from failing

Summary Week 9:
Shared individual code and started working on team function by first using a trial-and-error method to determine the position of each object and storing it in the program itself.
Date
Thursday November 11, 2025
Discussions
• Debated whether it's best to store the position or keep the list of python code for each object as separate objectives to be complete depending upon what the customer orders • How to increase the grip strength of the claw • How to ensure that the claws move with the arm and do not get stuck in one position • How to reduce the weight of the claw
Action items
• Re-printed the face plate with a huge hole in the middle, in-order to reduce the weight • Loosen the nuts and bolts just enough to ensure the claws do not get stuck and all 4 moves at the same time • Use rubber feet to increase grippiness of the claw

Summary Week 10:
Hand tested the final design and made any small adjustments as required. Printed both short and long attachments for the claw to determine the best length.

Date
Thursday November 18, 2025
Discussions
• How to ensure that the claw works consistently • What length of rod attachment for the motor should be used
Action items
• Test the final design with the Q arm itself • Make any adjustments needed to the position for each item with the end effector attached.
Summary Week 11:
Started working on the final project report. Worked on the summary as a team and then worked on each individual task for the appendices. Final testing and adjustments were made for the position of items.
Date
Thursday November 25, 2025
Discussions
• What should be included in the executive summary and how long should each sub-topic be. • What are some good last minute final adjustments to our claw • Who is going to comment and compile the python program
Action items
• Write the 1000-word executive summary and review it • Each team member needs to complete their respective tasks for the appendices • 3D print a wheel for the string • Comment and compile the final Python Code to submit before design studio interview
Summary Week 12:
Tested the picking up of objects for the first time using Q-Arm, got all objects till level 3.
Date
Wednesday December 3, 2025
Discussions
• What coordinates do we need to input to get to the individual objects • What gripper rotation to get the claw to properly open and close • Make hole for wheel bigger so that it fits on the end effector
Action items

- Individually tested using the console on pyzo to get exact coordinates for each object
- Trial and Error to get exact rotation for the gripper to operate properly
- Used screwdriver to make the hole big enough to fit on the end effector.

Appendix C: Comprehensive List of Sources

- [1] CADstudioCZ, "GitHub - CADstudioCZ/Autodesk-Inventor: Add-in tools for Autodesk Inventor," *GitHub*, 2025. <https://github.com/CADstudioCZ/Autodesk-Inventor>
- [2] "PLA Filament," *Filament 3D Printing | 3D4Makers.com*. <https://www.3d4makers.com/products/pla-filament>
- [3] "Paulin 3/16 x 1-1/2-inch Flat Head Square Drive Stove Bolt with Nut - Zinc Plated- 13 pcs | The Home Depot Canada," *The Home Depot Canada*, 2017. <https://www.homedepot.ca/product/paulin-3-16-x-1-1-2-inch-flat-head-square-drive-stove-bolt-with-nut-zinc-plated-13-pcs/1000120792>
- [4] "FindThisBest (UK)," *FindThisBest (UK)*, Jun. 17, 2021. <https://www.findthisbest.co.uk/brand/6077-cesfonjer>
- [5] "3D printed „Mechanical Crane Claw“ from bequ3[=]" www.youtube.com. <https://www.youtube.com/shorts/kZDDWaCRA0s>
- [6] *Avenue.cilmcmaster.ca*, 2025. <https://avenue.cilmcmaster.ca/d2l/le/lessons/712239/topics/5336250>
- [7] W3schools, "Python String Formatting," *W3schools.com*, 2022. https://www.w3schools.com/python/python_string_formatting.asp
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- [11] "7. Input and Output — Python 3.8.6 documentation," *docs.python.org*. <https://docs.python.org/3/tutorial/inputoutput.html>
- [12] Autodesk CAD Tutorials, "2D Engineering Drawings | Autodesk Inventor 2025 | Basics For Beginners | Step-by-Step," *YouTube*, Dec. 02, 2024. <https://www.youtube.com/watch?v=M9Q5H7vJyvQ>
- [13] "1/2 in. x 1 in. Clear Soft Rubber Like Plastic Self-Adhesive Rectangular Bumpers (8-Pack)," *Everbilt, The Home Depot Canada*. Accessed: Dec. 03, 2025. [Online]. Available: <https://www.homedepot.ca/product/everbilt-1-2-in-x-1-in-clear-soft-rubber-like-plastic-self-adhesive-rectangular-bumpers-8-pack-/1000847003>

Appendix D: Additional Documents

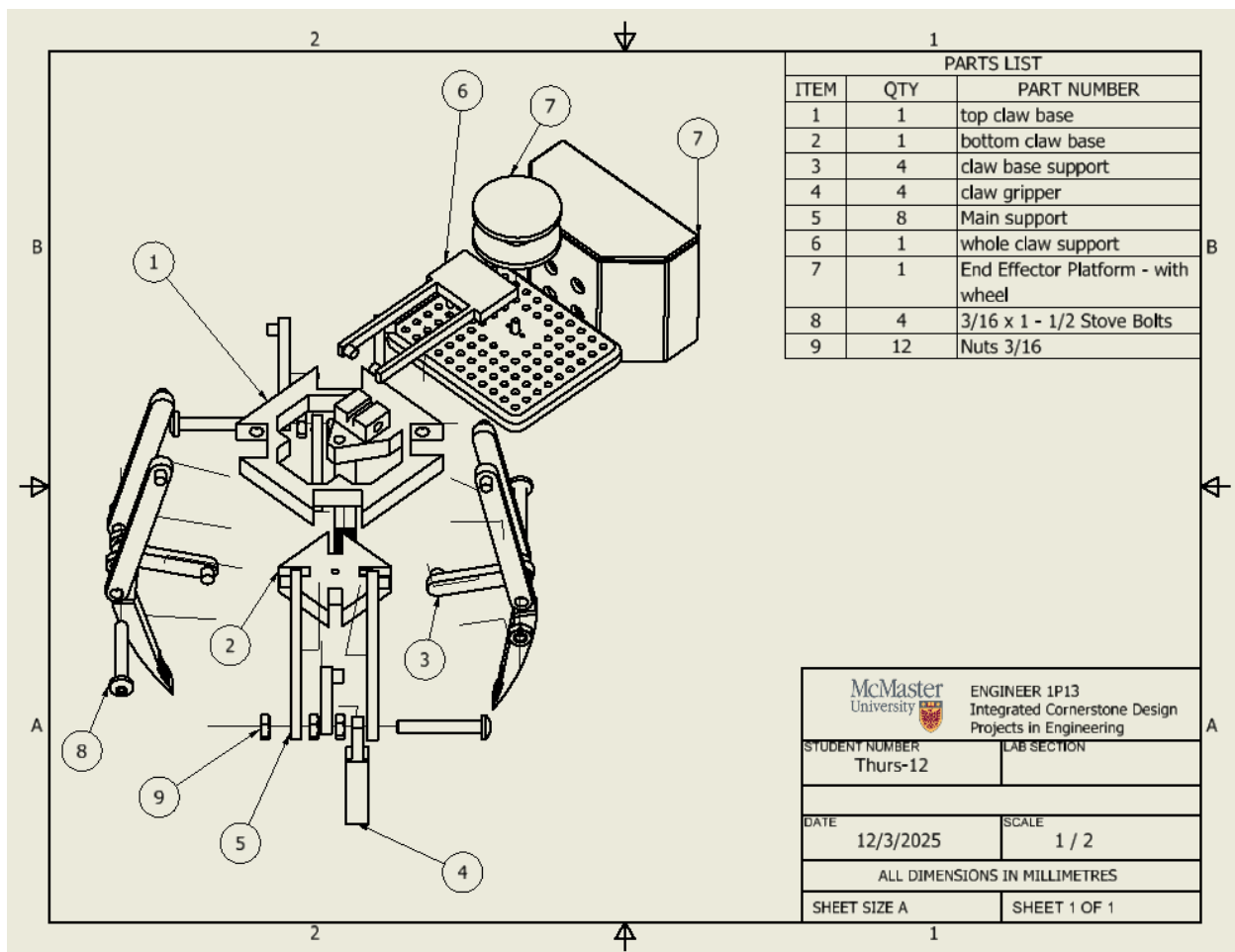


Fig 5: Exploded drawing of the claw assembly

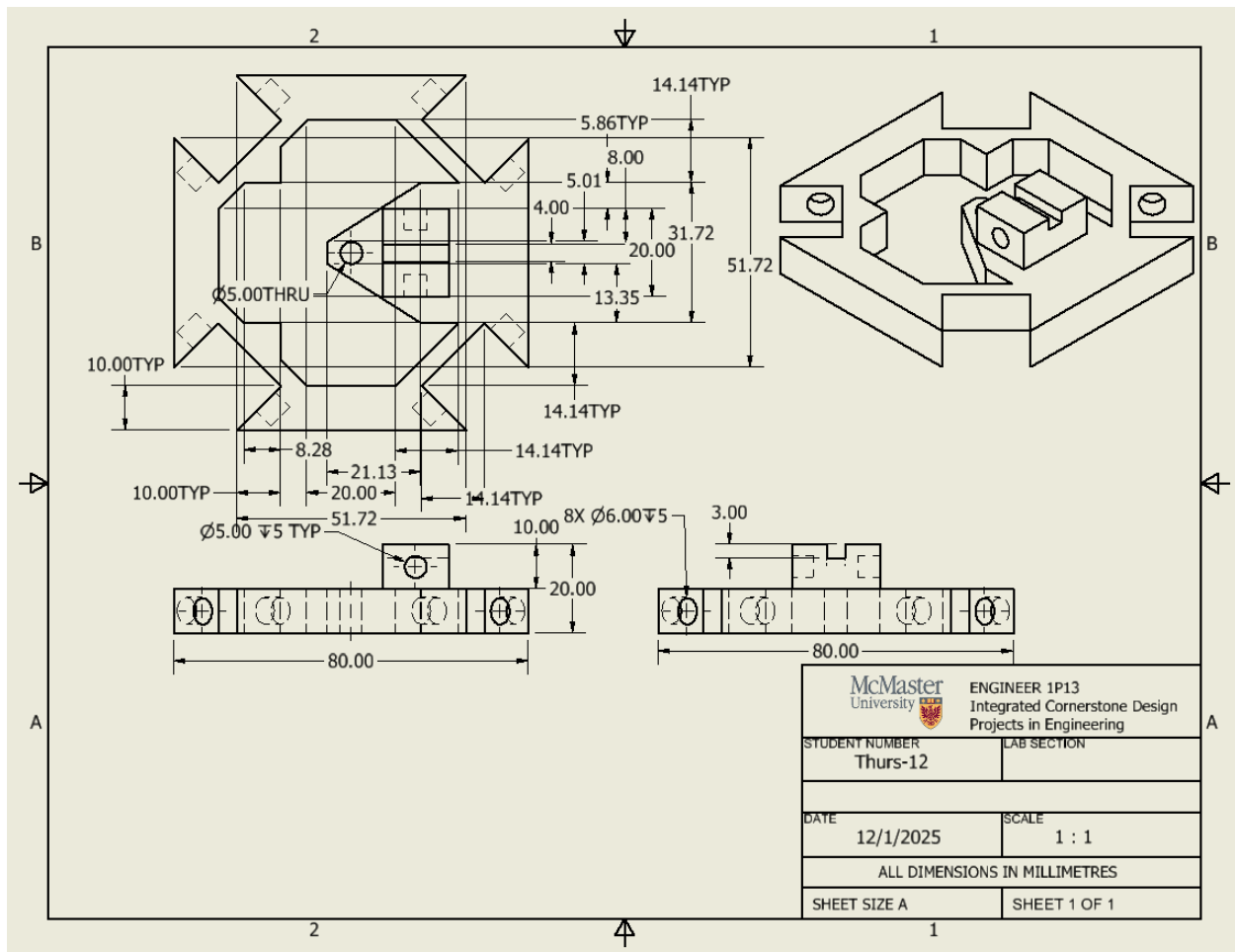


Fig 6: Engineering drawing of top plate of the claw

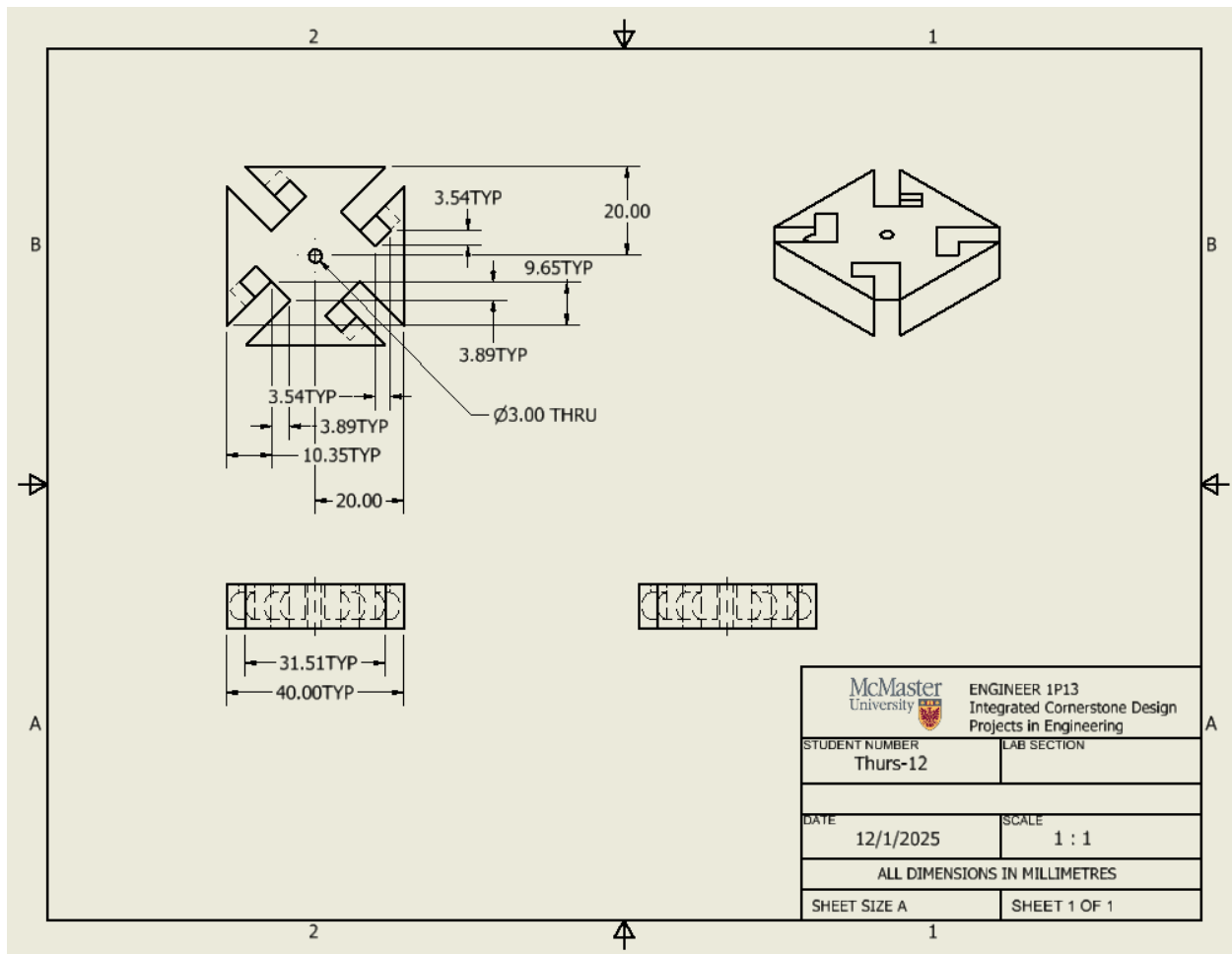


Fig 7: Engineering drawing of bottom plate

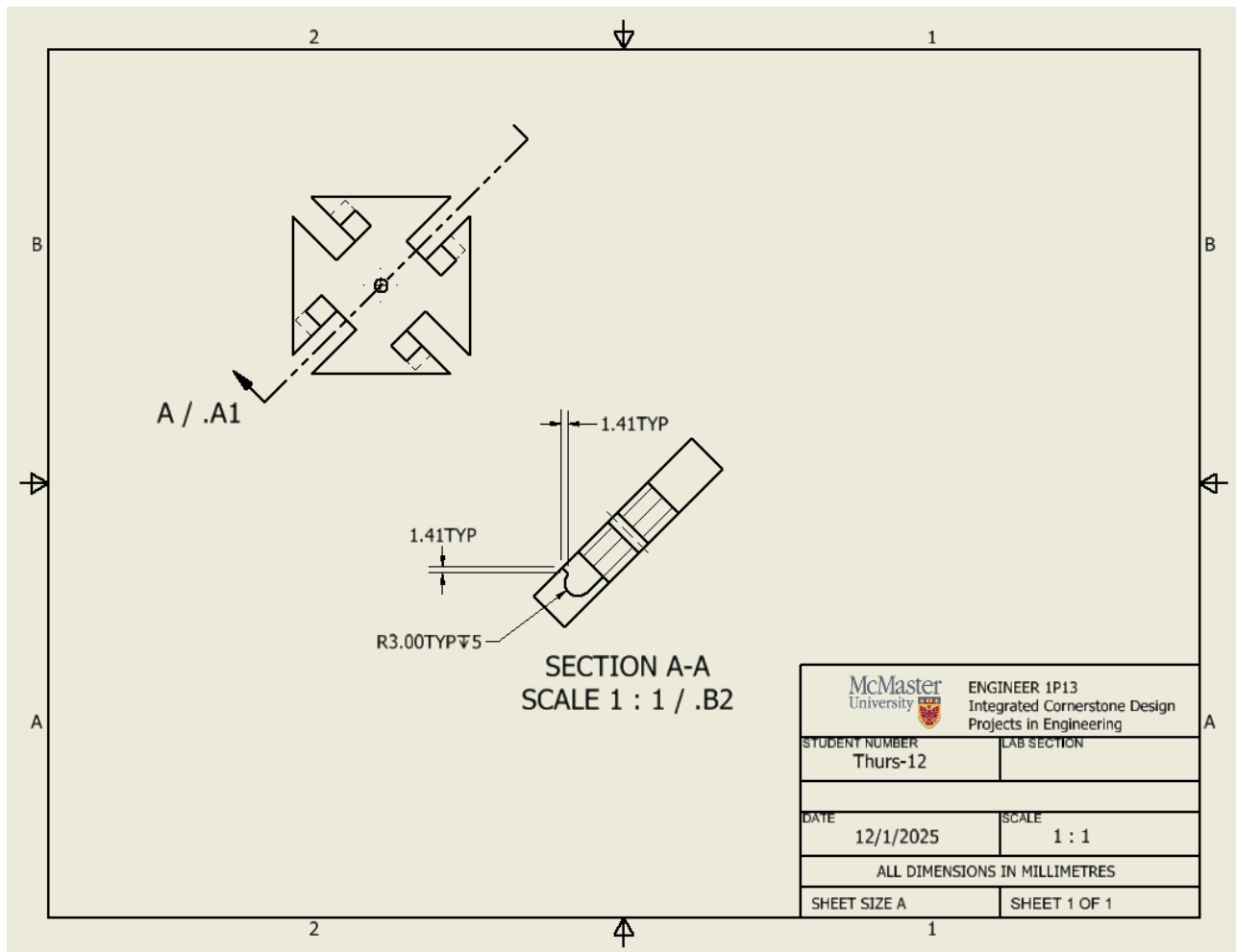


Fig 8: Engineering drawing of cross section of bottom plate

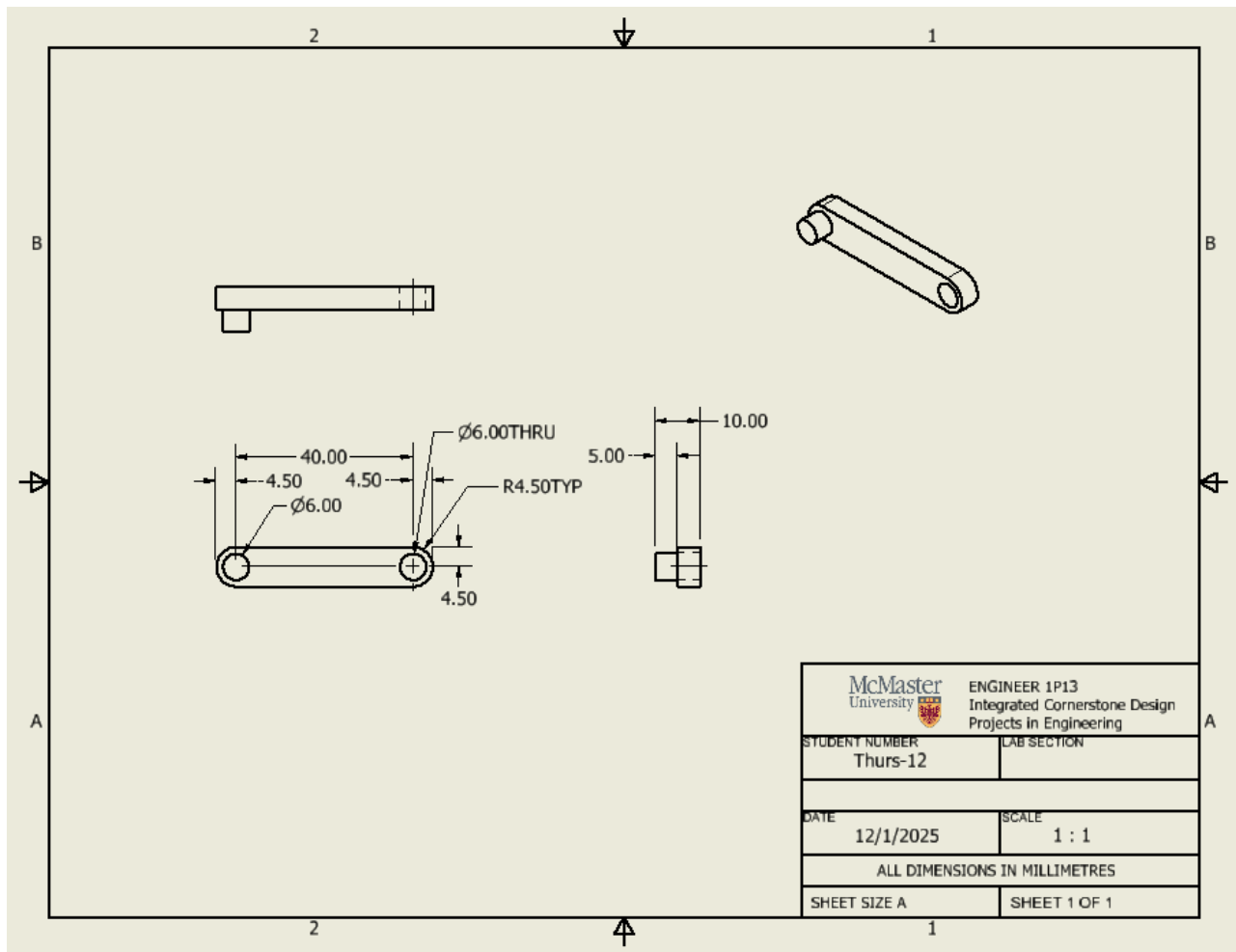


Fig 9: Engineering drawing of claw base support

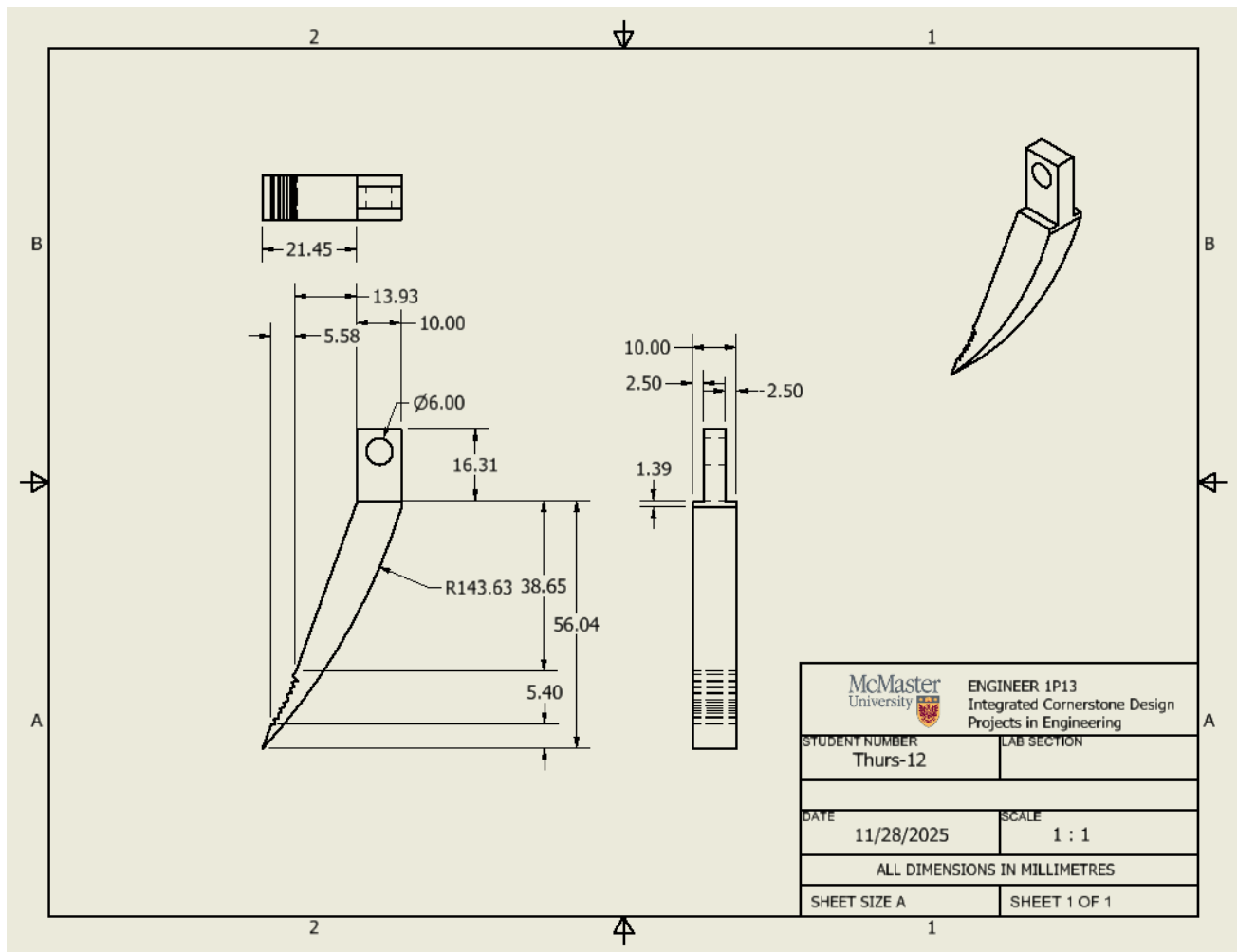


Fig 10: Engineering drawing of claw gripper

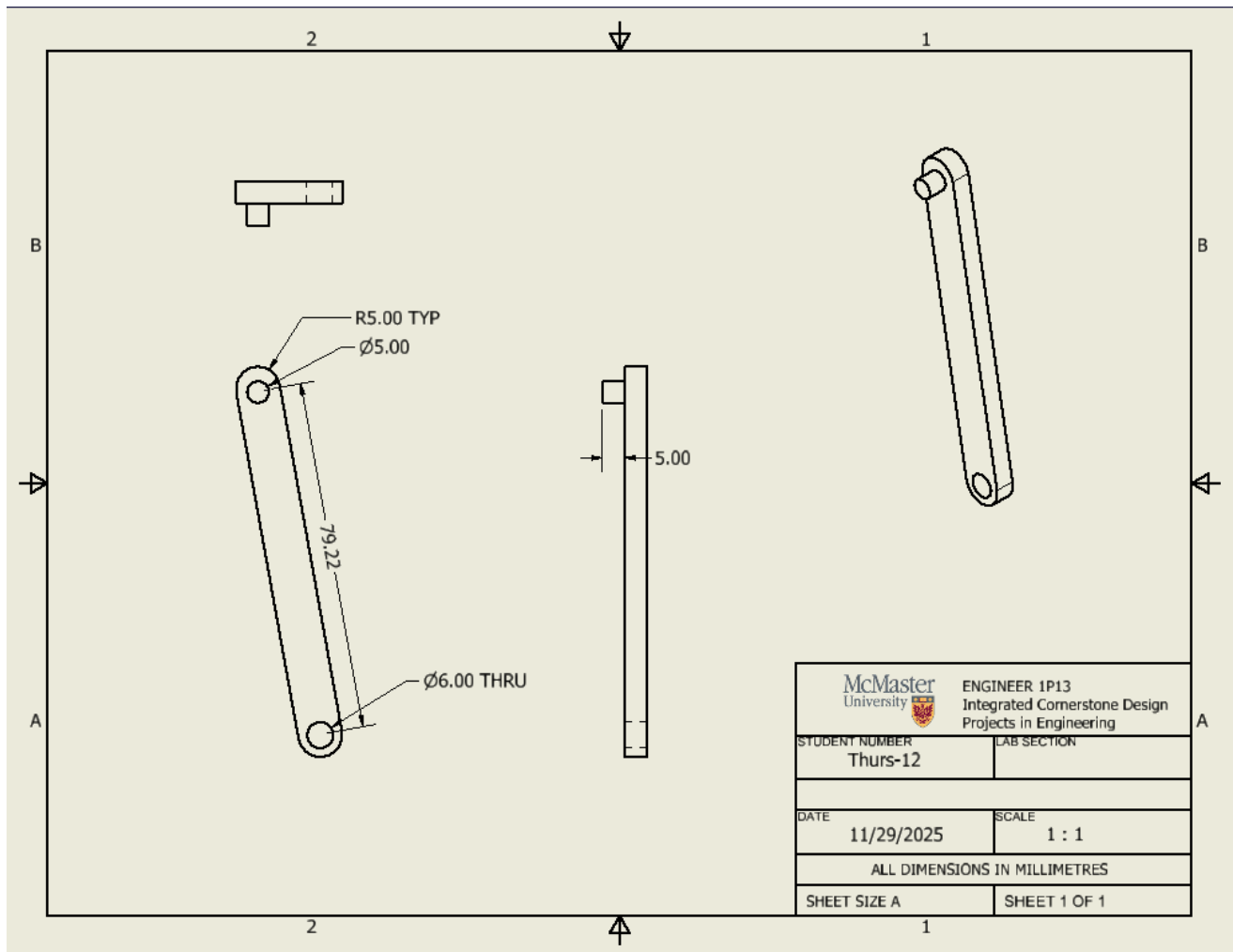


Figure 11: Engineering drawing of main support

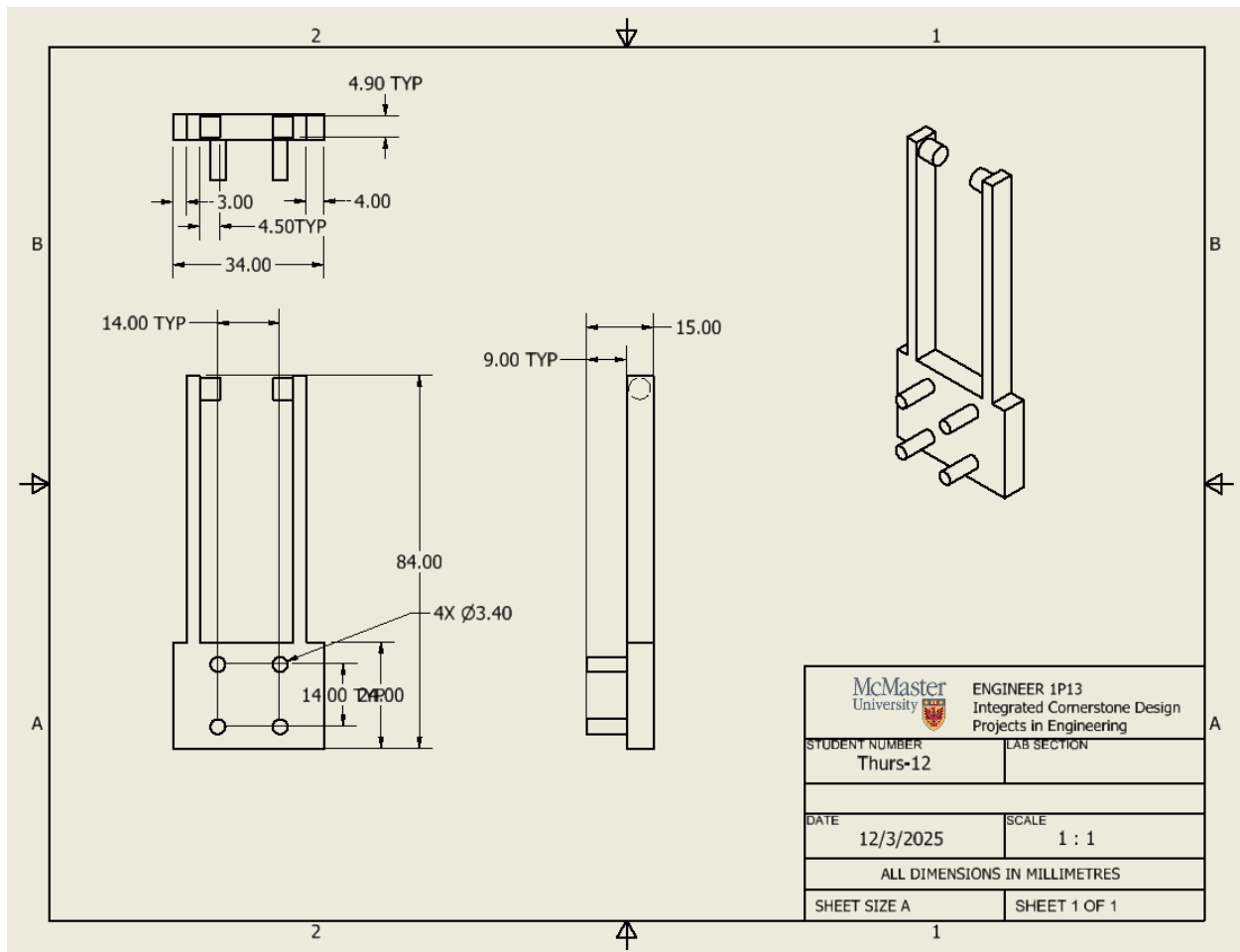


Figure 12: Engineering drawing of whole claw support

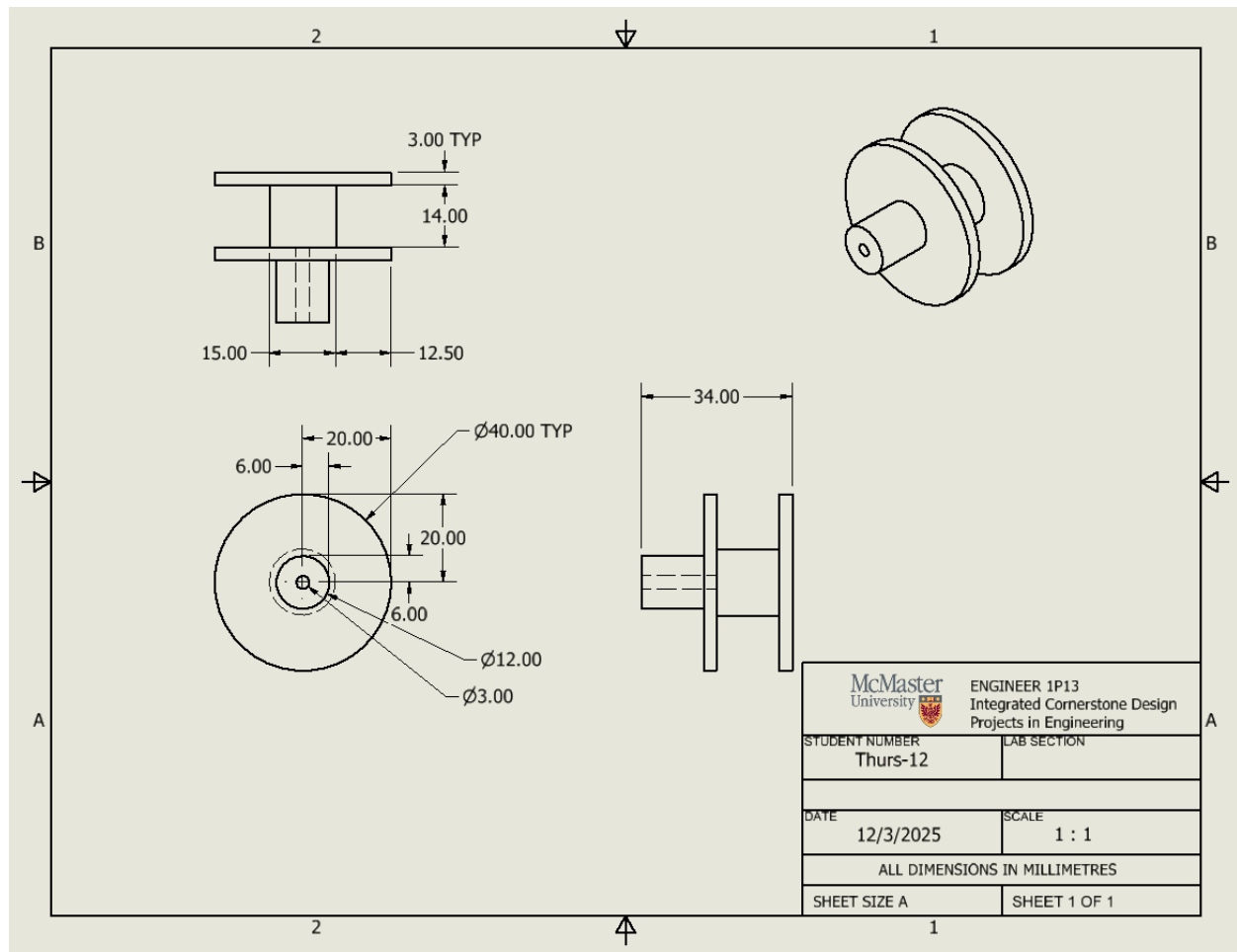


Figure 13: Engineering drawing of wheel

Appendix E: Design Studio Worksheets